

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
%pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2) (1.17.0)
```

```
dataset = pd.read_csv("2_letter_recognition.data", sep = ",")
```

```
pip install -U scikit-learn
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.8.0)
Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.10.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.13.1)
Requirement already satisfied: joblib>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.5.0)
```

```
X = dataset.iloc[:, 1 : 17]
Y = dataset.select_dtypes(include = [object])
```

```
X_train, X_validation, Y_train, Y_validation = train_test_split(X, Y, test_size = 0.20, random_state = 10)
```

```
scaler = StandardScaler()
scaler.fit(X_train)
```

```
▼ StandardScaler ⓘ ?
► Parameters
```

```
X_train = scaler.transform(X_train)
X_validation = scaler.transform(X_validation)
```

```
mlp = MLPClassifier(hidden_layer_sizes = (250, 300), max_iter = 1000000, activation = 'logistic')
```

```
%pip install yellowbrick
```

```
Requirement already satisfied: yellowbrick in /usr/local/lib/python3.12/dist-packages (1.5)
Requirement already satisfied: matplotlib!=3.0.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (3.9.0)
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (1.13.1)
Requirement already satisfied: scikit-learn>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (1.8.0)
Requirement already satisfied: numpy>=1.16.0 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (2.0.2)
Requirement already satisfied: cyclo in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (0.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (1.3.0)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (4.54.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (3.0.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (11.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.0.0,>=2.0.2) (2.9.0)
Requirement already satisfied: joblib>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.0.0) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.0.0) (3.5.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7) (1.17.0)
```

```

from sklearn.neural_network import MLPClassifier
from sklearn.metrics import confusion_matrix

# Create model
mlp = MLPClassifier(hidden_layer_sizes=(100,), max_iter=300)

# Train model
mlp.fit(X_train, Y_train)

# Predict
y_pred = mlp.predict(X_validation)

# Confusion matrix
cm = confusion_matrix(Y_validation, y_pred)

print(cm)

```

```

/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:1223: DataConversionWarning:
  y = column_or_1d(y, warn=True)
[[166  0  0  1  0  0  0  0  0  1  0  0  0  0  1  0  1  0
   0  0  0  0  0  0  0  0  0]
 [  0 137  0  0  1  0  0  1  1  0  1  0  0  1  0  0  0  1
   0  0  0  1  0  0  0  0]
 [  0  0 151  0  2  0  5  0  0  0  0  0  0  0  2  0  0  0
   0  1  0  0  0  0  0  0]
 [  0  2  0 165  0  0  0  0  0  0  0  0  0  2  3  0  1  1
   0  0  0  0  0  0  0  0]
 [  0  0  2  0 139  0  1  0  0  0  1  2  0  0  0  0  1  0
   1  0  0  0  0  1  0  0]
 [  0  0  0  0  0  1 132  0  1  0  0  0  0  0  0  1  0  0
   0  2  0  0  0  0  0  0]
 [  0  1  1  0  2  0 133  0  0  0  0  1  1  0  0  1  0  0
   0  0  0  0  1  0  0  0]
 [  0  1  0  2  1  0  1 126  0  0  4  0  0  1  0  0  2  0
   0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  1  0  0 142  5  0  0  0  0  0  0  0
   0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  1 134  0  1  0  0  0  0  0  1
   1  1  0  0  0  0  1  0]
 [  0  0  0  0  0  0  0  1  0  0 135  0  0  0  0  0  0  0
   0  0  0  0  0  0  0  0]
 [  0  0  0  0  2  0  1  0  0  1  0 159  0  0  0  1  0  0
   2  0  0  0  0  1  0  0]
 [  0  0  0  0  0  0  1  0  0  0  0  0 164  0  1  0  0  0
   0  0  1  0  1  0  0  0]
 [  1  0  0  0  0  0  0  0  0  0  1  0  0 145  0  0  0  1
   0  1  0  0  0  0  0  0]
 [  0  0  0  1  0  0  0  0  0  0  0  0  1  0 150  0  0  2
   0  0  0  0  0  0  0  0]
 [  1  1  0  0  0  4  0  1  1  0  0  0  0  0  0 150  0  0
   0  0  0  0  0  1  0]
 [  0  1  0  0  0  0  1  1  0  0  0  0  0  0  3  1 135  0
   0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  1  4  0  0  2  1  0  2  0  1  0 150
   0  0  0  0  0  0  0]
 [  1  0  0  0  2  1  0  0  0  0  0  0  0  0  0  0  1  0
  140 0  0  0  0  0  0]
 [  0  0  0  1  0  0  0  1  0  0  1  0  0  0  0  0  0  0
   0 157  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  1  1  0  0  0  0
   0  0 158  1  0  0  0]
 [  0  1  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0
   0  0  0 158  0  0  1]
 [  0  0  0  0  0  0  2  1  0  0  0  0  0  0  0  0  0  0
   0  0  0  0 155  0]
 [  0  0  0  0  0  1  0  0  0  1  2  0  0  0  0  0  1  0
   0  0  0  0  0 158  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0
   0  2  0  2  0  0 162]
 [  0  0  0  0  4  0  0  0  1  0  0  0  0  0  0  0  0  0
   2  0  0  0  0  0 141]]
/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:785: ConvergenceWarning:
  warnings.warn(

```

```
!pip install yellowbrick
```

Requirement already satisfied: yellowbrick in /usr/local/lib/python3.12/dist-packages (1.5)
 Requirement already satisfied: matplotlib!=3.0.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (3.8.0)
 Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (1.11.0)
 Requirement already satisfied: scikit-learn>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (1.3.2)
 Requirement already satisfied: numpy>=1.16.0 in /usr/local/lib/python3.12/dist-packages (from yellowbrick) (1.26.4)
 Requirement already satisfied: cycycler>=0.10.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.11.0)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.1.1)
 Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.53.0)
 Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.5)
 Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (24.1)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (10.4.0)
 Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.1.2)
 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0)
 Requirement already satisfied: joblib>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.4.2)
 Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.5.0)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil) (1.16.0)

```
mlp.fit(X_train, Y_train.values.ravel())
```

/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:785: ConvergenceWarning:
 warnings.warn(

MLPClassifier ⓘ ?		
Parameters		
<u>hidden_layer_sizes</u>		(100,)
<u>activation</u>		'relu'
<u>solver</u>		'adam'
<u>alpha</u>		0.0001
<u>batch_size</u>		'auto'
<u>learning_rate</u>		'constant'
<u>learning_rate_init</u>		0.001
<u>power_t</u>		0.5
<u>max_iter</u>		300
<u>shuffle</u>		True
<u>random_state</u>		None
<u>tol</u>		0.0001
<u>verbose</u>		False
<u>warm_start</u>		False
<u>momentum</u>		0.9
<u>nesterovs_momentum</u>		True
<u>early_stopping</u>		False
<u>validation_fraction</u>		0.1
<u>beta_1</u>		0.9
<u>beta_2</u>		0.999
<u>epsilon</u>		1e-08
<u>n_iter_no_change</u>		10
<u>max_fun</u>		15000

```
mlp.score(X_validation, Y_validation)
```

0.9605

```
predictions = mlp.predict(X_validation)
predictions
```

```
array(['C', 'I', 'H', ..., 'R', 'B', 'W'], dtype='<U1')
```

```
print("Accuracy: ", accuracy_score(Y_validation, predictions))
```

```
Accuracy:  0.9605
```

Start coding or [generate](#) with AI.